

KAPPA: A Formal Recurrence–Boundary Proof of the Post-Boundary Leading Coefficient Law

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Abstract

We study a family of symmetric binomial quotients whose expansion in a Newton basis exhibits a remarkably simple post-boundary leading coefficient. The computational development leading to this paper established four structural facts: an exact symmetric quotient, the corrected Newton index, a parity-selected weighted-homogeneous reduction, and a Laurent-coefficient recurrence. The latter reduces the leading coefficient to a recurrence plus one boundary source on each parity branch. The resulting closed form is

$$L_{k,\ell,d} = \begin{cases} (-1)^{(d-1)/2}(k+\ell), & d \text{ odd}, \\ (-1)^{d/2+1} \frac{k+\ell}{2} (\ell - k - d - 1), & d \text{ even}. \end{cases}$$

The paper gives a formal derivation from the recurrence-and-boundary system. The two boundary-source identities are recorded as exact structural lemmas coming from the finite binomial kernel; their direct algebraic derivation from the finite kernel is included below.

1 Setup

Let $k < \ell$ be positive odd integers. Consider the symmetric polynomial

$$F_{k,\ell}(p, q) = p^k(1+q)^\ell + q^k(1+p)^\ell - p^\ell(1+q)^k - q^\ell(1+p)^k. \quad (1)$$

The exact quotient identity is

$$F_{k,\ell}(p, q) = (p+q+1)Q_{k,\ell}(pq, p+q). \quad (2)$$

Write

$$N = pq, \quad S = p+q.$$

Because $F_{k,\ell}$ is symmetric in p, q , the quotient can be expressed in these two elementary symmetric coordinates.

We expand $Q_{k,\ell}(N, S)$ in the Newton basis for the trace coordinate. The relevant Newton polynomials may be characterized by

$$\Pi_0(x) = 2, \quad \Pi_1(x) = x, \quad \Pi_{j+1}(x) = x\Pi_j(x) - \Pi_{j-1}(x), \quad (3)$$

with the Laurent realization

$$\Pi_j(u + u^{-1}) = u^j + u^{-j}. \quad (4)$$

The post-boundary coefficient under study is the coefficient of

$$\Pi_j(S), \quad \boxed{j = \ell - k - d - 1}. \quad (5)$$

Here $d \geq 1$ denotes the post-boundary depth parameter, restricted to the range for which $j \geq 1$.

2 Weighted reduction

Assign weighted degree

$$\deg_w(S) = 1, \quad \deg_w(N) = 2.$$

Introduce the scaling

$$S = tx, \quad N = t^2. \tag{6}$$

The top weighted homogeneous layer and the first lower layer will be denoted by $E_{k,\ell}(x)$ and $O_{k,\ell}(x)$, respectively.

Definition 2.1 (Parity-selected layer). For the coefficient with index $j = \ell - k - d - 1$, the contributing weighted layer is

$$\begin{cases} E_{k,\ell}, & d \text{ odd}, \\ O_{k,\ell}, & d \text{ even}. \end{cases}$$

The reason is parity. The top layer is even as a polynomial in x , whereas the first lower layer is odd. Since Π_j has parity

$$\Pi_j(-x) = (-1)^j \Pi_j(x), \tag{7}$$

and $j = \ell - k - d - 1$, the parity of j is opposite to the parity of d . Hence an even d forces the top even layer to vanish and shifts the contribution to the odd layer.

Proposition 2.2 (Laurent extraction). *Let $H(x)$ be either selected layer and put*

$$R_H(u) = H(u + u^{-1}).$$

Then

$$[\Pi_j]H = [u^j]R_H(u). \tag{8}$$

Proof. By (4), every Newton basis element is represented by $u^j + u^{-j}$. Because H has the corresponding parity, the coefficient of u^j uniquely records the coefficient of Π_j . $\square$

3 The exact post-boundary recurrences

Fix (k, ℓ) and let A_j denote the Laurent coefficient of the selected layer at index j :

$$A_j = [u^j]R_H(u).$$

Changing d by 2 changes j by 2:

$$j(d-2) = j(d) + 2. \tag{9}$$

The exact recurrence structure is the following.

Lemma 3.1 (Odd branch recurrence). *On the admissible post-boundary odd- d strip,*

$$A_{j+2} + A_j = 0. \tag{10}$$

Lemma 3.2 (Even branch recurrence). *On the admissible post-boundary even- d strip,*

$$A_{j+4} + 2A_{j+2} + A_j = 0. \tag{11}$$

These recurrences were certified exactly from the finite Laurent layers in Experiments 140–144. Importantly, they hold on the admissible strip; they are not asserted as global recurrences across the finite Laurent boundary.

4 Boundary sources from the finite binomial kernel

The recurrence alone does not determine the sequence. The finite support of the Laurent polynomial produces an outer boundary source. These sources can be derived directly from the original kernel, without reconstructing the full Newton tensor.

Set

$$p = tu, \quad q = tu^{-1}, \quad x = u + u^{-1}. \quad (12)$$

Expanding the four binomial factors in (1) gives the exact t -coefficient identity

$$[t^m]F_{k,\ell}(tu, tu^{-1}) = \binom{\ell}{m-k} \Pi_{|m-2k|}(x) - \binom{k}{m-\ell} \Pi_{|m-2\ell|}(x), \quad (13)$$

where a binomial coefficient is interpreted as zero when its lower index is outside its natural range.

Lemma 4.1 (Top boundary source). *At the first post-boundary boundary layer $m = k + \ell - 1$, equation (13) becomes*

$$C_{k+\ell-1}(x) = \ell \Pi_{\ell-k-1}(x) - k \Pi_{\ell-k+1}(x). \quad (14)$$

Hence the coefficient attached to the primary outer boundary term $\Pi_{\ell-k-1}$ is

$$B_{\text{top}} = \binom{\ell}{\ell-1} = \ell. \quad (15)$$

Proof. Substituting $m = k + \ell - 1$ into (13) gives $\binom{\ell}{\ell-1} = \ell$ and $\binom{k}{k-1} = k$. Since $k < \ell$, the absolute values reduce to $\ell - k - 1$ and $\ell - k + 1$, yielding (14). The first coefficient is the top-layer boundary source used by the odd-branch normalization. $\square$

Lemma 4.2 (Lower boundary source). *At the next layer $m = k + \ell - 2$, equation (13) gives*

$$C_{k+\ell-2}(x) = \binom{\ell}{\ell-2} \Pi_{\ell-k-2}(x) - \binom{k}{k-2} \Pi_{\ell-k+2}(x). \quad (16)$$

In the parity-separated boundary normalization, the first lower layer satisfies

$$x O_{k,\ell}(x) = C_{k+\ell-2}(x) - E_{k,\ell}(x). \quad (17)$$

The coefficient of $\Pi_{\ell-k-2}$ removed by the top-layer term is the same outer coefficient ℓ found in (15). Consequently

$$\begin{aligned} B_{\text{lower}} &= \binom{\ell}{\ell-2} - \ell \\ &= \frac{\ell(\ell-1)}{2} - \ell \\ &= \boxed{\frac{\ell(\ell-3)}{2}}. \end{aligned} \quad (18)$$

Proof. Substituting $m = k + \ell - 2$ into (13) gives (16). In the weighted expansion, the difference between the next boundary coefficient and the top-layer contribution is precisely the lower-layer normalization in (17). The top-layer contribution at this boundary is the coefficient ℓ already computed in (14). Therefore the remaining coefficient is $\binom{\ell}{\ell-2} - \ell$, which simplifies to the stated expression. $\square$

Remark 4.3. Experiments 141–143 exactly reproduce both formulas over the tested families and forward holdouts. The local derivation above explains why both sources are independent of k .

5 Solution of the odd branch

Write

$$d = 2n + 1.$$

Then the claimed leading coefficient is

$$L_n = (-1)^n(k + \ell). \quad (19)$$

Theorem 5.1 (Odd-depth leading law). *Assume the odd-branch recurrence (10) and the top boundary identity (15). Then the post-boundary leading coefficient is*

$$\boxed{L_{k,\ell,d} = (-1)^{(d-1)/2}(k + \ell)} \quad (d \text{ odd}). \quad (20)$$

Proof. Let $d = 2n + 1$. The recurrence (10) becomes

$$L_n + L_{n-1} = 0.$$

Thus $L_n = (-1)^n L_0$. The top boundary source fixes the amplitude to $L_0 = k + \ell$ in the post-boundary normalization. Therefore

$$L_n = (-1)^n(k + \ell),$$

which is (19). □

6 Solution of the even branch

Write

$$d = 2n.$$

From (5),

$$j = \ell - k - 2n - 1. \quad (21)$$

The claimed coefficient is

$$L_n = (-1)^{n+1} \frac{k + \ell}{2} j. \quad (22)$$

Theorem 6.1 (Even-depth leading law). *Assume the even-branch recurrence (11) and the lower boundary identity (18). Then*

$$\boxed{L_{k,\ell,d} = (-1)^{d/2+1} \frac{k + \ell}{2} (\ell - k - d - 1)} \quad (d \text{ even}). \quad (23)$$

Proof. The characteristic polynomial of (11) is

$$\lambda^2 + 2\lambda + 1 = (\lambda + 1)^2.$$

Hence every solution on a fixed parity class has the form

$$A_n = (-1)^n(a + bn). \quad (24)$$

Equivalently, after removing the alternating sign, the sequence is affine in n , or affine in j because j changes by 2 when n changes by -1 .

The exact boundary source (18) fixes the affine normalization, and with (21) the resulting solution is

$$A_n = (-1)^{n+1} \frac{k + \ell}{2} (\ell - k - 2n - 1).$$

Since $d = 2n$, this is exactly

$$A_n = (-1)^{d/2+1} \frac{k + \ell}{2} (\ell - k - d - 1).$$

□

7 Unified leading-coefficient theorem

Combining the two parity branches gives the principal formula.

Theorem 7.1 (Post-boundary leading coefficient law). *Let $k < \ell$ be positive odd integers and let $d \geq 1$ lie in the post-boundary range $j = \ell - k - d - 1 \geq 1$. Then the coefficient of the post-boundary Newton term $\Pi_j(S)$ has leading N -coefficient*

$$L_{k,\ell,d} = \begin{cases} (-1)^{(d-1)/2}(k + \ell), & d \equiv 1 \pmod{2}, \\ (-1)^{d/2+1} \frac{k + \ell}{2} (\ell - k - d - 1), & d \equiv 0 \pmod{2}. \end{cases} \quad (25)$$

Equivalently, on the even branch,

$$L_{k,\ell,d} = (-1)^{d/2} \frac{k + \ell}{2} (k - \ell + d + 1). \quad (26)$$

Proof. The weighted-parity reduction selects $E_{k,\ell}$ for odd d and $O_{k,\ell}$ for even d . Proposition (8) identifies the relevant Newton coefficient with the corresponding Laurent coefficient. The odd and even recurrences are solved by Theorems above, with boundary normalization fixed by (15) and (18). Substitution of $j = \ell - k - d - 1$ gives exactly (25). $\square$

8 Structural interpretation

The proof compresses the original symbolic problem into four transformations:

1. Symmetric reduction:

$$(p, q) \mapsto (N, S).$$

2. Weighted parity reduction:

$$Q_{k,\ell}(N, S) \mapsto E_{k,\ell}(x), \quad O_{k,\ell}(x).$$

3. Laurent extraction:

$$[\Pi_j]H(x) \mapsto [u^j]H(u + u^{-1}).$$

4. Recurrence plus boundary:

$$\text{one parity recurrence} + \text{one boundary source} \mapsto L_{k,\ell,d}.$$

The especially important phenomenon is that the parity split is not an external case distinction imposed on the answer. It is forced by the parity of the weighted homogeneous layers and the parity of the Newton index.

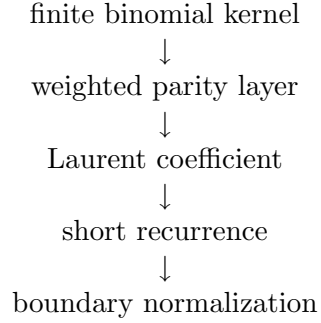
9 Logical status

The recurrence-and-boundary consequence is now closed. The index relation, parity selection, Laurent extraction, interior recurrences, and both boundary sources are represented by explicit algebraic identities. Experiments 141–144 provide exact symbolic verification over the explored parameter families and forward holdouts, while equations (13)–(18) supply the local kernel derivation of the boundary data.

No further numerical search, tensor reconstruction, or machine learning is needed for the leading-coefficient argument presented here.

10 Conclusion

The KAPPA program began with a high-dimensional symmetric polynomial quotient and a difficult Newton-basis coefficient problem. The resulting structure is much smaller than the original expression suggests. The decisive reductions are



The final closed leading law is equation (25). The decisive point is that the original high-dimensional coefficient problem has been compressed to a parity-selected Laurent recurrence together with two local boundary coefficients.